

AeroPulse BRICS

Cross-Border AI Climate Intelligence & Industrial Source Attribution Network

A federated planetary climate action platform fusing citizen smartphone photos and local IoT sensors with satellite Earth observation and meteorological wind vectors to detect hidden polluters, forecast 48-hour trans-boundary smog dispersion, and coordinate bilateral emergency response.

The Blind Spot: Atmospheric Physics vs. Sovereign Borders

Air pollution doesn't respect political boundaries. Over 300 million citizens across BRICS economic corridors face severe seasonal health emergencies due to three critical systemic failures:

1. SATELLITE TEMPORAL LAG

Earth observation satellites (Sentinel-5P, MODIS) only pass 1–2 times daily and are completely blind under heavy cloud cover, rain, or thick winter inversion fog.

2. HYPER-LOCAL INVISIBILITY

Government monitoring stations are sparse and clustered in city centers, completely missing midnight industrial soot dumps, rural brick kilns, and border stubble fires.

3. THE BLAME GAME & SECRECY

Neighboring states and nations trade accusations because neither has verifiable proof of where plumes originated, and domestic laws prohibit exporting raw telemetry.

AeroTrace: Reverse Plume Triangulation & Source Attribution

Pairing Google Gemini 2.0 Flash Multimodal Vision with Lagrangian atmospheric back-trajectory physics:

1. GEMINI MULTIMODAL VISION

Analyzes citizen sky photos to classify particulate type (Industrial Coal Soot vs. Biomass vs. Chemical mist) and calculates optical opacity percentage and plume rise.

2. LAGRANGIAN BACK-TRAJECTORY

Numerically inverts wind vectors backward in time against live meteorology, computing exact **plausible distance travelled (18.4 km)** and **transit time (1.25 hours)**.

3. UPWIND SECTOR FILTER

Constructs an upstream dispersion cone ($\pm 18^\circ$) querying registered industrial directories to rank candidate culprit factories with transparent, explainable AI reasoning.

Five Core Pillars of the AeroPulse Platform

48h

FORWARD PLUME FORECAST

2D Advection-Diffusion solver giving downwind schools & hospitals a 24-hour advance buffer.

1,420

BORDER FLUX KG/H

Virtual border line-integrals quantifying net cross-border pollutant transfer rates in real-time.

5 / 5

BRICS FEDERATED NODES

India, South Africa, Brazil, China, Russia training via FedAvg with zero raw data export.

CAP

OASIS V1.2 PROTOCOL

Automated bilateral alerts to upwind regulators and downwind public health centers.

How Google Gemini 2.0 Flash Powers AeroPulse

Deep, end-to-end integration across the multimodal data lifecycle:

MULTIMODAL SMOKE INSPECTION

Uses gemini-2.0-flash via the official google-genai SDK to inspect ambient sky photos, estimating optical extinction and distinguishing coal soot from steam.

EXPLAINABLE SOURCE REASONING

Synthesizes back-trajectory distance, wind alignment, and facility fuel profiles to output structured, evidence-backed attribution ranking reports for regulatory authorities.

MULTILINGUAL EMERGENCY ALERTS

Dynamically translates critical OASIS CAP advisories into BRICS native languages (Hindi, Portuguese, Chinese, Russian) for instantaneous community dissemination.

Ready for Planetary Climate Action

PROJECT DELIVERABLES

- **Full-Stack Working Prototype:** FastAPI + React WebGL
- **Automated Test Suite:** 12/12 pytest unit tests passing
- **Full HD Video Walkthrough:** 1080p MP4 in repository
- **Complete Documentation:** Architecture, Math, APIs in docs/
- **Open Source Code:** MIT Licensed on GitHub

SUBMISSION REPOSITORY & DEMO

- **GitHub Repository:** <https://github.com/githubvin/aeropulse>
- **Demo Video:** docs/aeropulse_demo.mp4
- **Hackathon Track:** BRICS Theme (Sustainability & Cooperation)
- **Google AI:** Gemini 2.0 Flash (google-genai SDK)